

A Practical Guide to Chain of thought reasoning

TOPIC -- CHAIN OF THOUGHT REASONING

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MIPRO (Multi-prompt Instruction Proposal Optimizer) optimizes the instructions and few-shot demonstrations of multi-stage language model programs, proposing small changes to module prompts and retaining those that improve a downstream performance metric without access to module-level labels or gradients. GEPA (Genetic-Pareto) is a reflective prompt optimizer for compound AI systems that combines language-model-based analysis of execution traces and textual feedback with a Pareto-based evolutionary search over a population of candidate systems; across four tasks, GEPA reports average gains of about 10% over reinforcement-learning-based Group Relative Policy Optimization (GRPO) and over 10% over the MIPROv2 prompt optimizer, while using up to 35 times fewer rollouts than GRPO. Open-source frameworks such as DSPy and Opik expose these and related optimizers, allowing prompt search to be expressed as part of a programmatic pipeline rather than through manual trial and error.

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Using language models to generate prompts LLMs themselves can be used to compose prompts for LLMs. The automatic prompt engineer algorithm uses one LLM to beam search over prompts for another LLM:

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Context engineering Context engineering is a related process that focuses on the context elements that accompany user prompts, which include system instructions, retrieved knowledge, tool definitions, conversation summaries, and task metadata. Context engineering is performed to improve reliability, provenance and token efficiency in production LLM systems. The concept emphasises operational practices such as token budgeting, provenance tags, versioning of context artifacts, observability (logging which context was supplied), and context regression tests to ensure that changes to supplied context do not silently alter system behaviour.

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A text-to-image prompt commonly includes a description of the subject of the art, the desired medium (such as digital painting or photography), style (such as

hyperrealistic or pop-art), lighting (such as rim lighting or crepuscular rays), color, and texture. Word order also affects the output of a text-to-image prompt. Words closer to the start of a prompt may be emphasized more heavily.

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Image prompting In 2023, Meta's AI research released Segment Anything, a computer vision model that can perform image segmentation by prompting. As an alternative to text prompts, Segment Anything can accept bounding boxes, segmentation masks, and foreground/background points.

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Chain-of-thought (CoT) prompting is a technique that allows large language models (LLMs) to solve a problem as a series of intermediate steps before giving a final answer. In 2022, Google Brain reported that chain-of-thought prompting improves reasoning ability by inducing the model to answer a multi-step problem with steps of reasoning that mimic a train of thought. Chain-of-thought techniques were developed to help LLMs handle multi-step reasoning tasks, such as arithmetic or commonsense reasoning questions. When applied to PaLM, a 540 billion parameter language model, according to Google, CoT prompting significantly aided the model, allowing it to perform comparably with task-specific fine-tuned models on several tasks, achieving state-of-the-art results at the time on the GSM8K mathematical reasoning benchmark. It is possible to fine-tune models on CoT reasoning datasets to enhance this capability further and stimulate better interpretability. As originally proposed by Google, each CoT prompt is accompanied by a set of input/output examples—called exemplars—to demonstrate the desired model output, making it a few-shot prompting technique. However, according to a later paper from researchers at Google and the University of Tokyo, simply appending the words "Let's think step-by-step" was also effective, which allowed for CoT to be employed as a zero-shot technique.

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There are two LLMs. One is the target LLM, and another is the prompting LLM. Prompting LLM is presented with example input-output pairs, and asked to generate instructions that could have caused a model following the instructions to generate the outputs, given the inputs. Each of the generated instructions is used to prompt the target LLM, followed by each of the inputs. The log-probabilities of the outputs are computed and added. This is the score of the instruction. The highest-scored instructions are given to the prompting LLM for further variations. Repeat until some stopping criteria is reached, then output the

highest-scored instructions. CoT examples can be generated by LLM themselves. In "auto-CoT", a library of questions are converted to vectors by a model such as BERT. The question vectors are clustered. Questions close to the centroid of each cluster are selected, in order to have a subset of diverse questions. An LLM does zero-shot CoT on each selected question. The question and the corresponding CoT answer are added to a dataset of demonstrations. These diverse demonstrations can then added to prompts for few-shot learning.

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Artist styles Some text-to-image models are capable of imitating the style of particular artists by name. For example, the phrase in the style of Greg Rutkowski has been used in Stable Diffusion and Midjourney prompts to generate images in the distinctive style of Polish digital artist Greg Rutkowski. Famous artists such as Vincent van Gogh and Salvador Dalí have also been used for styling and testing.

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Using gradient descent to search for prompts In "prefix-tuning", "prompt tuning", or "soft prompting", floating-point vectors are searched directly by gradient descent to maximize the log-likelihood on outputs. An earlier result uses the same idea of gradient descent search, but is designed for masked language models like BERT, and searches only over token sequences, rather than numerical vectors. Formally, it searches for $\arg \max_{\tilde{X}} \sum_i \log \Pr[Y^i | X \sim * \tilde{X}^i]$ where $X \sim$ ranges over token sequences of a specified length.

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Automated prompt generation Recent research has explored automated prompt engineering, using optimization algorithms to generate or refine prompts without human intervention. These automated approaches aim to identify effective prompt patterns by analyzing model gradients, reinforcement feedback, or evolutionary processes, reducing the need for manual experimentation.

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In-context learning A model's ability to temporarily learn from prompts is known as in-context learning. In-context learning is an emergent ability of large language models. It is an emergent property of model scale, meaning that breaks in scaling laws occur, leading to its efficacy increasing at a different rate in larger models than in smaller models. Unlike training and fine-tuning, which

produce lasting changes, in-context learning is temporary. Training models to perform in-context learning can be viewed as a form of meta-learning, or "learning to learn".